

Towards Optimized Mars Terrain Reconstruction Methods for Virtual Reality

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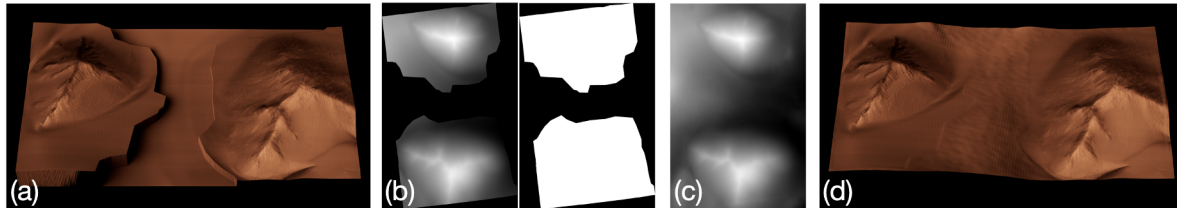


Figure 1: Example of Martian terrain restoration, performed using our proposed diffusion-based approach. a) A degraded terrain presents ill-formed geometrical features when rendered in 3D. b) Inputs of the restoration process: the terrain heightmap (left), and a binary mask (right) used to discriminate between valid (white) and invalid (black) regions. c) Output of our method; all invalid pixels have been restored. d) The reconstructed terrain presents structurally consistent features when rendered in 3D.

ABSTRACT

Virtual Reality enables the exploration of realistic outer space simulations, where accurate extraterrestrial terrain models are essential features. Due to extreme data retrieval challenges, such models often contain missing values that can hinder the 3D scenarios. We present a surface reconstruction method based on generative diffusion models, and compare it with other void-filling techniques in the task of restoring degraded Martian terrains. Reconstruction outputs were quantitatively assessed by comparing their error and structural consistency with the original values. Our method achieved the best performance on most metrics, suggesting promising prospects for extraterrestrial terrain visualization.

Index Terms: Virtual reality, Artificial intelligence, Mars.

1 INTRODUCTION

Virtual Reality (VR) in the space exploration domain has been widely explored in numerous applications [3]. The opportunities offered by VR in this context enable complex scenarios, otherwise extremely challenging to reproduce, to be effectively visualized within virtual environments [5, 6]. The visualization of accurate extraterrestrial terrains is a crucial feature for such outer space simulations [1]. When displaying planetary surfaces, gaps in terrain datasets may arise due to acquisition or processing errors. Interpolation methods provide useful estimates [2], but their 3D representation generally lacks visual consistency. In this work, we analyzed interpolation-based void-filling methods for terrain reconstruction on Mars, a relevant use case in space studies. We hereby propose an alternative approach based on a generative deep learning model. We advocate that AI can be beneficial in space-related VR by supporting scientific research in data-driven tasks and enriching the overall user experience in virtual environments. Our method builds upon previous work on deep learning-based void-filling studies [4]. Such methods, however, require additional information for training and generation, which is generally scarce in extraterrestrial sce-

narios. Instead, our approach avoids using supplemental material other than the terrains themselves, making it suitable for operating in data-scarce contexts.

2 METHODS AND RESULTS

We performed terrain reconstruction using RePaint¹, an inpainting algorithm that leverages image-based generative models, specifically diffusion models. RePaint guides a diffusion model to fill the missing parts of a degraded image, while keeping the generation itself unconditional. “Unconditional” refers to a model that does not require additional prompts (such as images or text) for content generation, in contrast with conditional models. RePaint employs a binary mask in its computations, but the diffusion model itself does not, thus not constraining generation on supplementary information and avoiding overfitting in data-scarce extraterrestrial contexts. To use RePaint on Martian surfaces, we trained a Denoising Diffusion Probabilistic Model² (DDPM) from scratch to generate Martian-like terrains unconditionally. Training and inference were performed using terrains represented as single-channel, 32-bit floating-point 2D heightmaps, and then rendered as 3D surfaces. The training set was derived from the HiRISE³ terrain survey. Random crops of variable size, with each side spanning from 512 to 2048 pixels, were sampled from there to obtain a total of 12,000 elements. GDAL⁴ was used to read and process the original HiRISE survey. Training of the DDPM was conducted over 100 epochs, with a batch size of 16. The model resolution was set to 128×128 pixels; each training element was normalized in the range $[-1, 1]$ during runtime and resized using nearest-neighbor interpolation. The process took approximately 8 hours on 2 NVIDIA A40 GPUs. We conducted a preliminary test run of 1000 crops (different from the ones in the training set) randomly sampled from HiRISE. These crops contained only valid pixels, which served as ground truth for evaluation; synthetic binary masks were artificially generated. We compared our method with other void-filling and inpainting methods [2]: Inverse Distance Weighting (IDW), the Navier-Stokes (NS)

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¹<https://huggingface.co/docs/diffusers/main/en/api/pipelines/repaint>

²<https://huggingface.co/docs/diffusers/main/en/api/pipelines/ddpm>

³<https://www.uahirise.org/dtm/>

⁴<https://gdal.org/en/stable/>

	RMSE ↓	MAE ↓	PSNR ↑	EMD ↓	SSIM ↑	LPIPS ↓	FID ↓
<i>Ours</i>	0.0364 ± 0.0315	0.0261 ± 0.0218	39.4494 ± 7.8701	0.1366 ± 0.1390	0.9660 ± 0.0352	0.0754 ± 0.0322	11.0065
IDW	0.0455 ± 0.0256	0.0326 ± 0.0186	35.7504 ± 5.8163	0.1756 ± 0.1483	0.9620 ± 0.0352	0.1365 ± 0.0476	61.2217
NS	0.0438 ± 0.0258	0.0315 ± 0.0189	36.2110 ± 6.0938	0.1699 ± 0.1493	0.9650 ± 0.0275	0.1181 ± 0.0463	42.2967
KR	0.0412 ± 0.0426	0.0311 ± 0.0342	39.3492 ± 9.6646	0.1572 ± 0.1865	0.9684 ± 0.0354	0.0973 ± 0.0680	27.7074

Table 1: Average results of quantitative metrics for the test run (1000 samples). FID is not an average, but a single score for the entire test. Metrics were calculated on normalized samples (range [0, 1]). ↑: higher is better; ↓: lower is better. The best results for a metric are in bold.

algorithm, and kriging (KR) were considered. Reconstruction outputs were evaluated quantitatively (Table 1). We computed metrics that evaluate distribution error and value differences: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR), and Earth Mover’s Distance (EMD). We also computed image similarity metrics: Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and Fréchet Inception Distance (FID). We also gathered preliminary insights on structural and geometrical consistency of the reconstruction outputs by rendering them as 3D surfaces; some examples are shown in Figure 2. They were imported into Autodesk Maya 2024⁵, and rendered using the Arnold⁶ renderer to simulate diverse light and shadow conditions.

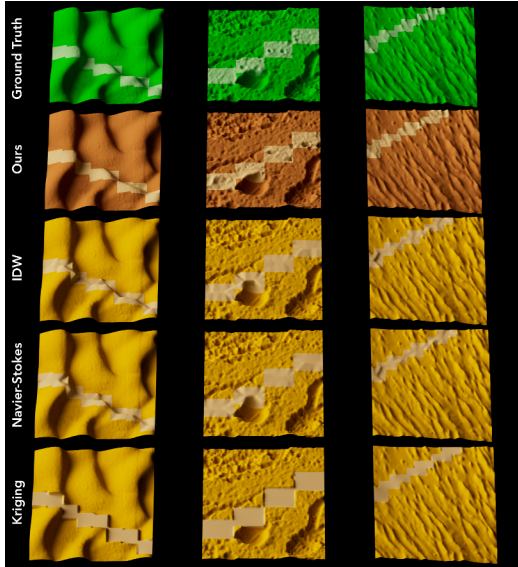


Figure 2: Visual assessment of results. Each column represents a different example. Each row represents outputs of the specified method, excluding the first (ground truth). Highlighted parts in each terrain represent the binary masks used.

3 DISCUSSION

The comparative test showed that our method outperformed other interpolation-based techniques across all error-based metrics, showing the ability to generate terrains that closely match the original values. Kriging ranked second, achieving close results due to its statistical nature, which derives correlations from valid pixels and provides additional information. IDW and Navier-Stokes produced faultier outputs. Regarding similarity-based metrics, kriging achieved the best SSIM score. This result may be due to SSIM’s structural comparison component, which can penalize reconstructions that are structurally coherent yet misaligned (ours) in favor

of flatter, less detailed restorations (kriging). LPIPS and FID indicated our method as the best-performing. Visual assessment was critical for contextualizing results in terms of their ability to recreate realistic environments. Kriging achieved numerically close performances to our method, and sometimes higher; however, the rendered surfaces showed largely improved structural and geometrical consistency in the latter compared to the former. Our approach generated complex patterns, whereas kriging mostly produced flat surfaces (Figure 2). IDW and Navier-Stokes also created geometric artifacts and discrepancies. In general, the presented method was the best data processing technique for visualization in VR. Current main limitations include inference times, which averaged 44 seconds on the described hardware, posing challenges for use in real-time applications; moreover, comparative assessments with other deep learning-based methods across various terrestrial and extraterrestrial datasets are lacking. Future work will focus on improving the neural architecture and the inpainting algorithm. Furthermore, user studies will be conducted to determine whether the reconstructed terrains are perceived as plausible by users and usable by experts (astronauts, scientists) in real-world space exploration scenarios. Such studies could provide insights into the effectiveness of our method and the broader context of how generated 3D content is perceived in VR.

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REFERENCES

- [1] G. L. Catalano, E. Topa, and A. M. Soccini. *Integrating Terrain Data into Virtual Reality Systems for Outer Space Exploration*, p. 221–229. Springer Nature Switzerland, 2024. doi: 10.1007/978-3-031-71707-9_16 1
- [2] S. Crema, M. Llena, A. Calsamiglia, J. Estrany, L. Marchi, D. Vericat, and M. Cavalli. Can inpainting improve digital terrain analysis? comparing techniques for void filling, surface reconstruction and geomorphometric analyses. *Earth Surface Processes and Landforms*, 45(3):736–755, january 2020. doi: 10.1002/esp.4739 1
- [3] J. Pirkker. The potential of virtual reality for aerospace applications. In *2022 IEEE Aerospace Conference (AERO)*, p. 1–8. IEEE, Mar. 2022. doi: 10.1109/aero53065.2022.9843324 1
- [4] K. Shih-Huang Lo and J. Peters. Diff-dem: A diffusion probabilistic approach to digital elevation model void filling. *IEEE Geoscience and Remote Sensing Letters*, 21:1–5, 2024. doi: 10.1109/lgrs.2024.3403835 1
- [5] A. M. Soccini, M. Marelli, N. Balossino, C. Bar, V. Basso, M. Luceteforte, D. Perlo, F. Racca, and L. Rocci. Virtual Reality Interface for Multidisciplinary Physical Analysis of Space Vehicles. In *EuroVR 2014 - Conference and Exhibition of the European Association of Virtual and Augmented Reality*. The Eurographics Association, 2014. doi: 10.2312/eurovr.20141330 1
- [6] A. M. Soccini, M. Marelli, N. Balossino, and V. Basso. ‘ixv-trajectory’ and ‘ixv-asset’: Virtual reality applications for the aerothermodynamics analysis of ixv. In *2015 IEEE Virtual Reality (VR)*, p. 397–398. IEEE, march 2015. doi: 10.1109/vr.2015.7223463 1

⁵<https://www.autodesk.com/products/maya/overview>

⁶<https://www.autodesk.com/products/arnold/overview>